

- 3 Mammals such as sheep, *Ovis aries*, and goats, *Capra hircus*, are important agricultural animals that are sometimes kept together in mixed flocks. Very occasionally, live offspring are born from a mating between a male sheep and a female goat.

In sheep $2n = 54$ and in goats $2n = 60$.

- (a) (i) Calculate the diploid chromosome number of the hybrid offspring of a sheep and a goat.

..... [1]

- (ii) Outline why the classification of sheep and goats suggests that hybridisation between them should **not** be likely to occur.

.....
..... [1]

- (b) Normal (wild-type) goats have a gold and black coat colour pattern, known as bezoar, and are also horned (have horns). Domestic goats may have a white coat and may be hornless (do not have horns).

These variations are coded for by two unlinked genes:

- white coat colour, coded for by the dominant allele of the gene **A/a**
- hornless, coded by the dominant allele of the gene **H/h**.

A cross between a white hornless goat and a bezoar horned goat produced offspring of four different phenotypes.

Draw a genetic diagram to show the genotypes of the two parents, their gametes and the offspring, and the phenotypes of the offspring.

[4]

- (c) Horns on agricultural animals such as goats and cattle can be dangerous to the farmer and to other animals. Horns are often prevented from growing in 5-day-old animals by a stressful procedure called disbudding.

Genetic modification can cause a deletion in the allele **h** coding for horns in cattle embryos, so that the allele no longer codes for a functional protein and the embryos grow into cattle that are hornless.

- (i) State an **ethical** advantage of this example of genetic modification.

.....
..... [1]

- (ii) Suggest why genetic modification that causes a deletion in the horned allele, in established breeds of dairy cattle, is preferable to selective breeding for hornless animals.

.....
.....
..... [1]